Chapter 5 Joint Probability Distributions and Random Sample

5.1 Jointly Distributed Random Variables

The Joint Probability Mass Function for Two Discrete Random Variables

The joint probability mass function $p(x, y)$:

$$p(x, y) = P(X = x, Y = y)$$

For a set A consisting of pairs of (x, y) values, to obtain $p[(X, Y) \in A]$,

$$p[(X, Y) \in A] = \sum \sum_{(x,y) \in A} p(x, y)$$

Two requirements for a pmf

$$p(x, y) \geq 0, \sum_x \sum_y p(x, y) = 1$$

The marginal Probability Mass function

The marginal probability mass function of X and Y, denoted by $p_X(x)$ and $p_Y(y)$, respectively, are given by

$$p_X(x) = \sum_y p(x, y), p_Y(y) = \sum_x p(x, y)$$

The Joint Probability Density Function for Two Continuous Random Variables

$f(x, y)$ is the joint probability density function for X and Y if for any two-dimensional set A

$$P[(X, Y) \in A] = \int \int_A f(x, y) dx dy$$

Two requirements for a joint pdf

- $f(x, y) \geq 0$
- $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1$

If A is the two-dimensional rectangle set $\{(x, y) : a \leq x \leq b, c \leq y \leq d\}$, then

$$P[(X,Y)=A]=P(a\le X\le b, c\le Y\le d)=\int_a^b\int_c^d f(x,y)\,dy\,dx$$

It is the volume under density surface above A.

Marginal Probability Density Function

The marginal probability density functions of X and Y, denoted by $f_X(x)$, $f_Y(y)$, respectively are given by

$$f_X(x)=\int_{-\infty}^{\infty} f(x,y)\,dy, -\infty<x<\infty$$
$$f_Y(y)=\int_{-\infty}^{\infty} f(x,y)\,dx, -\infty<y<\infty$$

Independent Random Variables

Two random variables X and Y are said to be independent if for every pair of x and y values,

$$p(x,y)=p_X(x)\cdot p_Y(y), (X,Y \text{ are discrete})$$
$$f(x,y)=f_X(x)\cdot f_Y(y), (X,Y \text{ are continuous})$$

More than Two Random Variables

For more random variables,

$$p(x_1,x_2,\cdots,x_n)=P(X_1=x_1,X_2=x_2,\cdots,X_n=x_n)(\text{discrete variables})$$
$$P(a_1\le X_1\le b_1,\cdots,a_n\le X_n\le b_n)=\int_{a_1}^{b_1}\cdots\int_{a_n}^{b_n} f(x_1,\cdots,x_n)\,dx_n\cdots dx_1(\text{continuous variables})$$

Multinomial Experiment

An experiment consisting of n independent and identical trials where each trial can result in any one of r possible outcomes. Let $p_i=P$(Outcome i on any particular trial), and define random variables by $X_i=\text{he number of trials resultin in outcome } i (i=1,2,\cdots,r)$, the joint pmf is called the multinomial distribution, shown to be

$$p(x_1,\cdots,x_r)=\frac{n!}{x_1!x_2!\cdots x_r!}p_1^{x_1}\cdots p_r^{x_r}, x_i=0,1,\cdots \text{ with } x_1+x_2+\cdots+x_r=n$$

Independent

For more random variables, they are said to be independent if for every pair, every triple and so on, the joint pmf or pdf of the subset is equal to the product of the marginal pmf or pdf.

Conditional Distribution

The conditional probability density function of Y given that $X=x$ is

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)}, -\infty < y < \infty$$

if X, Y are discrete, then

$$f_{Y|X}(y|x) = \frac{p(x,y)}{p_X(x)}, -\infty < y < \infty$$

5.2 Expected Values, Covariance and Correlation

The Expected Value of a function $h(x, y)$

The expected value of a function $h(x, y)$, denoted by $E[h(x, y)]$ or $\mu h(x, y)$ is

$$E[h(X,Y)] = \sum_x \sum_y h(x,y) \cdot p(x,y); X, Y \text{ are discrete}$$

and

$$E[h(X,Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x,y) f(x,y) \, dx \, dy; X, Y \text{ are continuous}$$

Covariance

When X and Y are not independent, it is frequency of interest to assess how strongly they are related to one another

The Covariance between two rv's X and Y is

$$Cov(X,Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

for discrete

$$\sum_X \sum_Y (x - \mu_X)(y - \mu_Y) p(x,y)$$

for continuous

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) f(x,y) \, dx \, dy$$

- If strong positive relationship, $Cov(x, y)$ should be quite positive
- If strong negative relationship, $Cov(x, y)$ should be quite negative
- If not strongly related, $Cov(x, y)$ near 0.

Proposition

$$Cov(X,Y) = E(XY) - \mu_X \mu_Y$$

Note that

$$Cov(X, X) = E(X^2) - \mu_X^2 = V(X)$$

Correlation

The correlation coefficient of X, Y, denoted by $Corr(X, Y)$, $_{X,Y}$ or just , is defined by

$$_{X,Y} = \frac{Cov(X, Y)}{\sigma_X \cdot \sigma_Y}$$

Proposition

If a and c are either both positive or both negative

$$Corr(aX + b, cY + d) = Corr(X, Y)$$

and for any two rv's X and Y, $-1 \le Corr(X, Y) \le 1$.

If X and Y are independent, then $= 0$, but $= 0$ does not imply independence. They are said to be uncorrelated.

$= 1 \text{ or } -1$ if and only if $Y = aX + b$ for some numbers a and b and $a \neq 0$.

5.3 Statistics and Their Distributions

From this section, we consider function of n random variables $X_1, \cdots, X_n$ focusing especially on their average $(X_1, X_2, \cdots, X_n)/n$. We call any such function, itself a random variable, a statistic. We studies the distribution of statistics, like sample mean, sample standard deviation and so on.

In summary, the value of the individual sample observations vary form sample to sample, so in general the value of any quantity computer from sample data, and the value of a sample characteristic used as an estimate of the corresponding population characteristic, will virtually never coincide with what is being estimated.

Statistic

A statistic is any quantity whose value can be calculated from sample data.

A statistic is a random variable since there is uncertainty as to what value of particular statistic will result before obtaining data. A statistic will be denoted by an uppercase letter. Calculated or observed value of the statistic uses lowercase letter.

The probability distribution of a statistic is sometimes referred to as its sampling distribution. It depends on

- The population distribution(normal, uniform, etc.)
- The sample size n

- The method of sampling(replacement or without replacement)

Random Sample

The rv's $X_1, X_2, \cdots, X_n$ are said to form a simple random sample of size n if

- The X_i's are independent rv's
- Every X_i has the same probability distribution

And we say that the X_i's are independent and identically distributed(i.i.d).

Sampling with replacement or from an infinite population is random sampling. Sampling without replacement from a finite population is generally considered not random sampling. However, if $n/N \leq 0.05$, it is approximately random sampling.

Deriving the Sampling Distribution of a Statistic

- Method 1: Calculations based on probability rules.
- Method 2: Carrying out a simulation experiments.

Example 5.20

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A large automobile service center charges \$40, \$45, and \$50 for a tune-up of four-, six-, and eight-cylinder cars, respectively. If 20% of its tune-ups are done on four-cylinder cars, 30% on six-cylinder cars, and 50% on eight-cylinder cars, then the probability distribution of revenue from a single randomly selected tune-up is given by

x	40	45	50
$p(x)$	0.2	0.3	0.5

$\mu = 46.5$

$\sigma^2 = 15.25$

Suppose on a particular day only two servicing jobs involve tune-ups.

Let X_1 = the revenue from the first tune-up &

X_2 = the revenue from the second,

which constitutes a random sample with the above probability distribution.

5.3 Statistics and Their Distributions

Example 5.20 (Cont')

x_1	x_2	$p(x_1,x_2)$	$\bar{x}$	$\bar{s}^2$
40	40	0.04	40	0
40	45	0.06	42.5	12.5
40	50	0.10	45	50
45	40	0.06	42.5	12.5
45	45	0.09	45	0
45	50	0.15	47.5	12.5
50	40	0.10	45	50
50	45	0.15	47.5	12.5
50	50	0.25	50	0

Known the Population Distribution

x	40	42.5	45	47.5	50
$p_x(x)$	0.04	0.12	0.29	0.30	0.25

$$\mu_{\bar{X}} = E(\bar{X}) = 46.5 = \mu$$

$$\sigma_{\bar{X}}^2 = V(\bar{X}) = \sum \bar{x}^2 \cdot P_{\bar{X}}(\bar{x}) - \mu_{\bar{X}}^2 = 7.625 = \frac{15.25}{2} = \frac{\sigma^2}{2}$$

The variance of $\bar{x}$ is precisely half that of the original variance (because n=2)

$\bar{X}$ sampling distribution is centered at the population mean μ and S^2 sampling distribution is centered at the population variance σ^2

Simulation Experiments

It is usually used when a derivation via probability rules is too difficult or complicated to be carried out. Such an experiment is virtually always done with the aid of a computer.
The following characteristics of an experiment must be specified

- The statistic of interest
- The population distribution
- The sample size n
- The number of replications k. (the actual sampling distribution emerges as $k \to \infty$)

When n increases, no matter what type the population distribution is, the $\bar{X}$ sampling distribution will close to normal distribution.

5.4 The Distribution of the Sample Mean

Proposition

Let $X_1, \cdots, X_n$ be a random sample from a distribution with mean value μ and standard deviation σ. Then

$$
\begin{aligned}
E(\bar{X}) &= \mu_{\bar{X}} = \mu \\
V(\bar{X}) &= \sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}, \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}
\end{aligned}
$$

With ${}_0 = X_1 + \cdots + X_n$(the sample total),

$$
\begin{aligned}
E({}_0) &= n\mu \\
V({}_0) &= n\sigma^2, \sigma_{{}_0} = \sqrt{n}\sigma
\end{aligned}
$$

Proposition

Let $X_1, \cdots, X_n$ be a random sample from a normal distribution with mean μ and standard deviation σ. Then for any n, $\bar{X}$ is normally distributed as is ${}_0$.

The Central Limit Theorem (CLT)

Let $X_1, \cdots, X_n$ be a random sample from any distribution with mean μ and variance σ^2, then if n is sufficiently large, $\bar{X}$ has approximately a normal distribution with

$$
\mu_{\bar{X}} = \mu, \sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}
$$

and $_0$ also has approximately a normal distribution with

$$\mu_0 = n\mu, \sigma_0^2 = n\sigma^2$$

Usually if $n > 30$, the CLT can be used.

5.5 The Distribution of a Linear Combination

Linear Combination

Given a collection of n random variables $X_1, \cdots, X_n$ and n numerical constants $a_1, \cdots, a_n$, the rv

$$Y = a_1X_1 + a_2X_2 + \cdots + a_nX_n = \sum_{i=1}^{n} a_iX_i$$

is called a linear combination of the X_i's.

Proposition

Let $X_1, \cdots, X_n$ have mean values $\mu_1, \cdots, \mu_n$ respectively and variances $\sigma_1^2, \cdots, \sigma_n^2$.

$$E\left(\sum_{i=1}^{n} a_iX_i\right) = \sum_{i=1}^{n} a_iE(X_i) = \sum_{i=1}^{n} a_i\mu_i$$

for any X_i.

If X_i are independent,

$$V\left(\sum_{i=1}^{n} a_iX_i\right) = \sum_{i=1}^{n} a_i^2V(X_i) = \sum_{i=1}^{n} a_i^2\sigma_i^2$$

and $\sigma_{\sum_{i=1}^{n} a_iX_i} = \sqrt{\sum_{i=1}^{n} a_i^2\sigma_i^2}$

For any X_i,

$$V\left(\sum_{i=1}^{n} a_iX_i\right) = \sum_{i=1}^{n}\sum_{j=1}^{n} a_ia_jCov(X_i, X_j)$$

Corollary

$$E(X_1 - X_2) = E(X_1) - E(X_2)$$

and if they are independent,

$$V(X_1 - X_2) = V(X_1) + V(X_2)$$

note that the sign is sum.

Proposition

If $X_1, \cdots, X_n$ are independent, normally distributed. then any linear combination of X_i's also has a normal distribution.